MLP Training Under Different Train/Test Splits on the Steel Plates Faults Dataset (Extension of Assignment 1 Baseline Study)

Nikita Chernysh
Faculty of Electrical Engineering and Informatics,
Technical University of Košice
nikita.chernysh@tuke.sk

Abstract

This report extends the previous study "Exploratory Data Analysis and Evaluation of Machine Learning Algorithms on the Steel Plates Faults Dataset" by introducing a simple Multi-Layer Perceptron (MLP) classifier trained on the same preprocessed Steel Plates Faults dataset. The objective is to evaluate how different train/test data splits (50/50, 60/40, 70/30, and 80/20) affect training behavior and generalization. For each split, the same fixed MLP architecture with ReLU activation is used; a validation set is reserved during training to monitor learning dynamics. Final performance is evaluated on the test set using accuracy, precision, recall, and F1-score. As a baseline, one traditional model from Assignment 1 (Gradient Boosting) is evaluated on the same test set to ensure a valid comparison.

Index Terms

MLP, Neural Networks, ReLU, Train/Test Split, Validation, Steel Plates Faults, Gradient Boosting, Classification.

I. Introduction

FAULT detection in steel plates is a critical quality control task in manufacturing, where accurate classification of defects ensures structural reliability and minimizes production costs. In Assignment 1, the Steel Plates Faults dataset was explored, preprocessed, and multiple traditional machine learning models were benchmarked. The main outcome was that Gradient Boosting achieved the strongest overall performance among the tested baselines.

This report serves as a direct continuation and extension of Assignment 1. We reuse the same dataset and preprocessing pipeline to keep the comparison objective and reproducible. The goal here is to train a simple MLP classifier (MLPClassifier) with at least one hidden layer and ReLU activation, and then systematically compare results across several train/test split ratios (50/50, 60/40, 70/30, and 80/20). During training, a validation subset is reserved to monitor learning dynamics. Finally, we evaluate test performance and compare it with the Assignment 1 baseline (Gradient Boosting) on the same test split.

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II. Dataset Overview

This work uses the *Steel Plates Faults* dataset (UCI Machine Learning Repository), consisting of 1,941 samples and 27 numerical features. The target variable represents seven fault classes: *Pastry*, *Z_Scratch*, *K_Scratch*, *Stains*, *Dirtiness*, *Bumps*, and *Other_Faults*. The class distribution is imbalanced, with *Other_Faults* being the majority class.

TABLE I
Target Class Distribution (Reference from Assignment 1)

Class	Count	Proportion (%)
Other_Faults	673	34.67
Bumps	402	20.71
K_Scratch	391	20.14
Z_Scratch	190	9.79
Pastry	158	8.14
Stains	72	3.71
Dirtiness	55	2.83

III. Data Preprocessing

To ensure a valid and objective comparison with Assignment 1, we reuse the same preprocessing strategy. The pipeline includes:

1) Removal of highly correlated features (threshold on absolute correlation; same criterion as Assignment 1).
2) Standardization of features using z-score scaling (`StandardScaler` fitted on training data only).
3) Stratified train/test splitting using a fixed random seed (seed = 5296) to ensure reproducibility.
4) Conversion of multi-output labels to a single categorical class label (via `idxmax`).
5) Computation of class weights (used for reference and imbalance awareness).

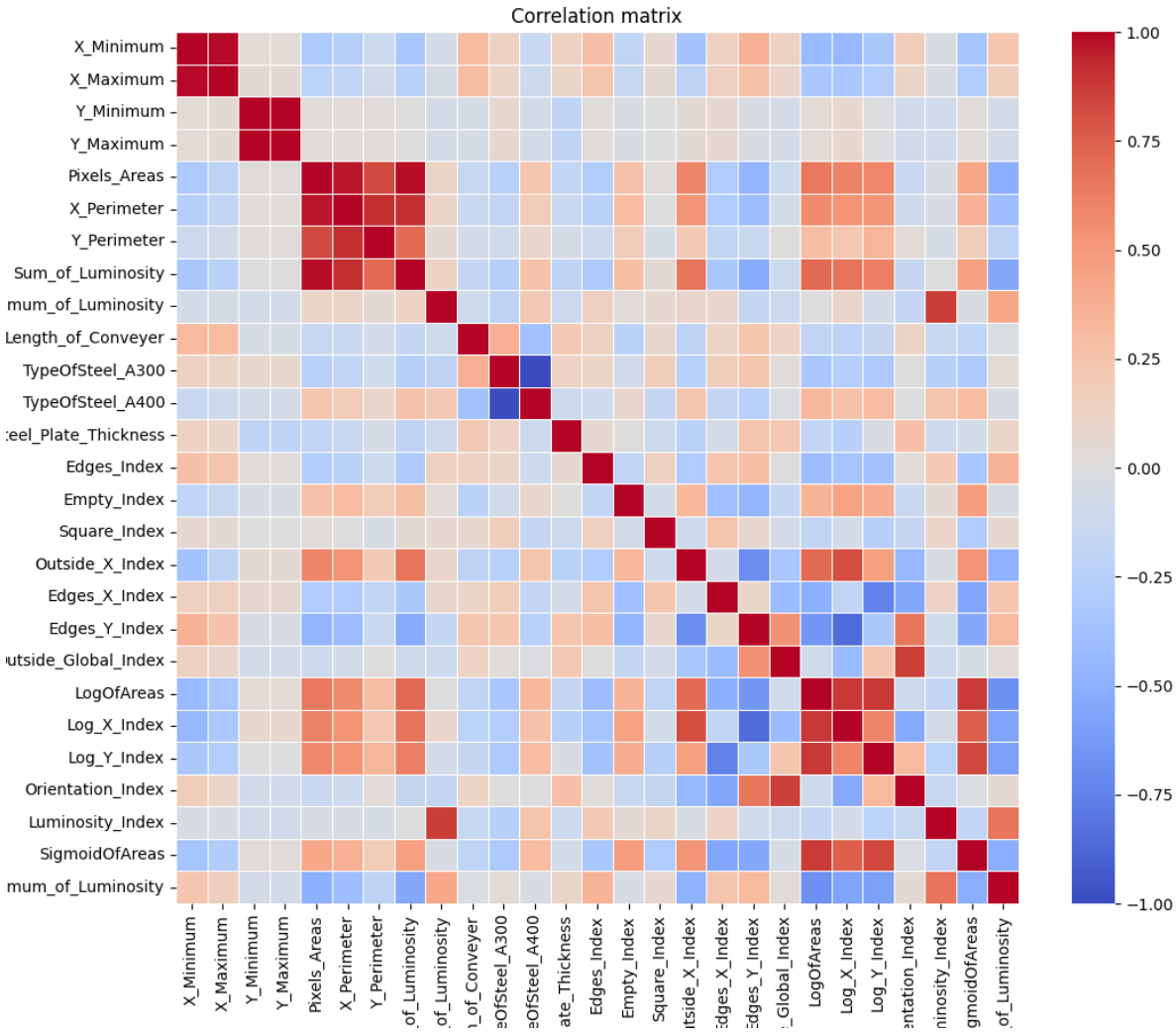

Fig. 1. Correlation analysis heatmap (consistent with Assignment 1). Highly correlated features were removed before modeling.

IV. Models

A. Baseline Model (Reference from Assignment 1)

As a baseline, we use the best-performing traditional algorithm from Assignment 1: **Gradient Boosting**. This baseline is evaluated on the same test split as the MLP to ensure that comparisons are valid. Baseline configuration is kept consistent with Assignment 1.

B. MLP Model

The core of this assignment is a simple MLP classifier trained on the same preprocessed dataset.

TABLE II
BASELINE MODEL CONFIGURATION (ASSIGNMENT 1 REFERENCE)

Model	Key Parameters
Gradient Boosting	200 estimators, learning_rate=0.1, max_depth=10

1) Fixed Architecture: To satisfy the assignment requirement (at least one hidden layer and ReLU activation) and to keep the experiments comparable, the MLP architecture is fixed across all splits:

- Hidden layers: (128, 64, 32)
- Activation: ReLU
- Output: 7-class classification (softmax implied by the loss used in `MLPClassifier`)

2) Training Hyperparameters: All experiments use the same training configuration:

- Solver/optimizer: Adam
- Learning rate: 1×10^{-3}
- L2 regularization: $\alpha = 1 \times 10^{-4}$
- Batch size: 32
- Maximum epochs: 200
- Validation set: reserved during training (fixed fraction of the training data)
- Early stopping: enabled (patience-based stopping)

TABLE III
MLP CONFIGURATION SUMMARY (FIXED FOR ALL SPLITS)

Component	Setting
Hidden layers	(128, 64, 32)
Activation	ReLU
Optimizer	Adam
Learning rate	1×10^{-3}
Regularization (L2)	$\alpha = 1 \times 10^{-4}$
Batch size	32
Max epochs	200
Validation split	20% of training set (during training)
Early stopping	enabled
Random seed	5296

V. EXPERIMENTAL SETUP

A. Train/Test Splits

We evaluate the MLP under four train/test splits required by the assignment:

- 50/50
- 60/40
- 70/30
- 80/20

For each split:

- the same preprocessing pipeline is applied,
- the same MLP architecture and hyperparameters are used,
- a validation subset is reserved from the training data to monitor learning.

B. Evaluation Metrics

We report standard classification metrics on the test set:

- Accuracy
- Precision
- Recall
- F1-score

C. Learning Curves

To analyze training behavior and generalization, we monitor training dynamics over epochs.

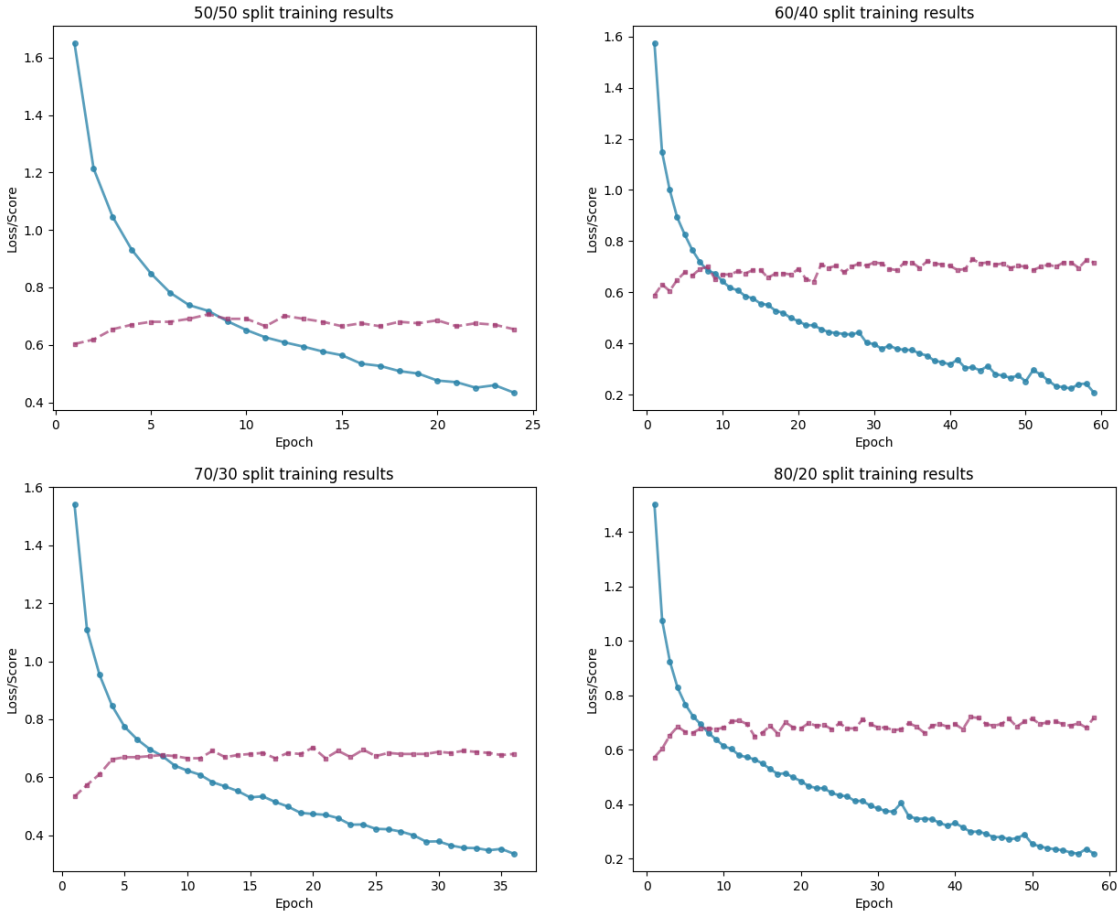

Fig. 2. Learning curves for train/test split.

VI. Results and Discussion

A. Test Performance Across Splits

Table IV reports the MLP test metrics for each train/test split and the baseline result. The baseline (Gradient Boosting) corresponds to Assignment 1 and is evaluated on the same test split as the MLP for 80/20 to ensure a valid comparison.

TABLE IV
Test Results for MLP Under Different Train/Test Splits and Baseline Comparison

Metric	50/50	60/40	70/30	80/20	Baseline
Accuracy	0.650875	0.697555	0.692967	0.704370	0.740360
Precision	0.658212	0.704854	0.695657	0.707517	0.747899
Recall	0.650875	0.697555	0.692967	0.704370	0.740360
F1-score	0.649044	0.699689	0.692610	0.704122	0.742469

B. Impact of Training Set Size

Overall, the MLP performance improves as the training set becomes larger. The 50/50 split shows the weakest results (accuracy 0.6509, F1 0.6490), while the best MLP performance is achieved at 80/20 (accuracy 0.7044, F1 0.7041). This trend is expected: with more training data, the network can learn more stable decision boundaries and generalize better.

The learning curves (Figures 2) should be used to interpret training stability and potential overfitting. In practice, smaller training sets typically lead to noisier validation behavior and earlier saturation.

C. Baseline Comparison

Despite improved performance with larger training sets, the MLP does not outperform the baseline Gradient Boosting model. On the shared 80/20 test split, the baseline achieves higher accuracy (0.7404) and F1-score (0.7425) compared to the MLP (accuracy 0.7044, F1 0.7041). This indicates that for this dataset and the fixed simple MLP configuration, Gradient Boosting remains a stronger choice.

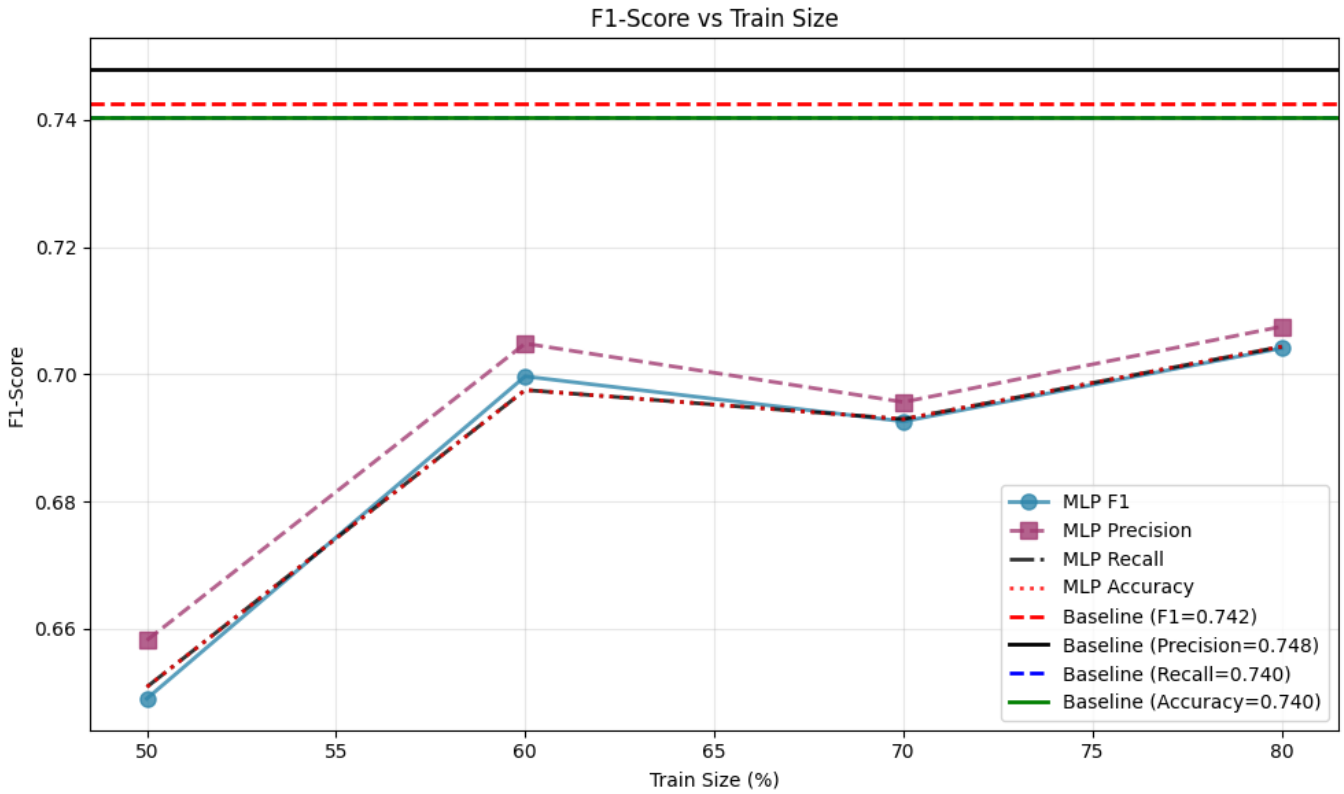

Fig. 3. Summary comparison of MLP performance across splits against the baseline model.

VII. Conclusion

This report extended Assignment 1 by training a simple ReLU-based MLP classifier on the same preprocessed Steel Plates Faults dataset and evaluating it under multiple train/test splits (50/50, 60/40, 70/30, and 80/20). The results confirm that increasing the training set size improves MLP generalization, with the best MLP performance achieved at 80/20 (accuracy 0.7044, F1 0.7041).

However, compared to the Assignment 1 baseline (Gradient Boosting) on the same test split, the MLP does not provide an improvement. The baseline achieves higher accuracy (0.7404) and F1-score (0.7425). Therefore, under the fixed "simple MLP" constraint of this assignment and without extensive hyperparameter tuning, Gradient Boosting remains the stronger model for this dataset.

Appendix A
Submitted Files

The submission includes:

- PDF report: `Testovanie_Chernysh.pdf`
- Jupyter notebook: `train.ipynb`
- Source code: `data_preprocessing.py`, `mlp_model.py`, `baseline_model.py`

References

[1] UCI Machine Learning Repository, "Steel Plates Faults Data Set," 2015. [Online]. Available: https://archive.ics.uci.edu/ml/datasets/steel+plates+faults
[2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.